Jose Ramon Chang Irias, Ph.D.

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PERSONAL PROFILE

I'm currently a Principal Engineer at Jabil's Chihuahua facility, where my responsibilities center on leveraging and enabling data to drive decisions across the manufacturing process. This includes production line monitoring, manufacturing intelligence tooling, and multiple Python based dashboards that I develop and maintain.

Previously, I was Senior Deputy Manager (AI R&D) at a publicly traded TWSE multinational, where I drove multi million dollar vendor partnerships to reduce storage energy consumption through AI driven optimization. I defined technical roadmaps, coached engineers, and aligned product, sales, and testing teams to deliver measurable performance gains across large scale storage clusters.

I hold a Ph.D. from National Cheng Kung University and specialize in deep learning and applied AI. My research spans neuroimaging, skin feature tracking, and the evaluation of large language models for scientific reproducibility, with multiple SCIE indexed publications and over 100 citations on Google Scholar. I have hands on experience across Mexico, Taiwan, Chile, and Sweden, and I am driven to combine academic rigor with real world impact through high performance AI solutions.

EDUCATION

Doctor of Philosophy, Mechanical Engineering
National Cheng Kung University, Tainan, Taiwan
DISSERTATION- Predictive modelling and deep learning for quantifying human health. November 2024

Master of Science, Mechanical Engineering
Kun Shan University, Tainan, Taiwan July 2017

Bachelor of Science, Double major in Mechanical and Electronics Engineering
Kun Shan University, Tainan, Taiwan July 2016

EXPERIENCE

Jabil March 2026 – Now
Chihuahua, Mexico (Principal Engineer)

- Used AI and ML to optimze the manufacturing processes of a major account.

Promise Technology Taiwan March 2025 – December 2025
Hsinchu, Taiwan (Senior Deputy Manager)

- Reported directly to CEO to establish and lead a team of three engineers to implement AI solutions to improve the throughput and energy saving of storage systems.
- Assisted the founder and general manager, James Lee, in establishing the AI R&D division by defining project ideas, designing the product roadmap, training of engineers and documenting research procedures.
- Led a multi-million dollar collaboration with Toshiba on applying AI for saving energy of large scale storage systems.

National Cheng Kung University June 2024 – September 2024
Tainan, Taiwan (Researcher)

- Carried out deep learning research on model compression using knowledge distillation enhanced by model explainability techniques to visualize the neural pathways of the network. This approach not only reduced the size of models but also made their decision-making process more transparent and easier to understand.

Altum Lab — July 2022 – May 2024
Santiago, Chile — (Lead data scientist)

Altum Lab is a Chilean startup focused on the improvement production processes with algorithms based on Artificial Intelligence.Their SaaS platform, is capable of reducing the use of energy resources, predicting the quality and characteristics of finished products according to their raw materials, and optimizing the use of machines in the operation.

- Ensure quality and integrity of the data science behind projects of the agricultural, mining, and aquaculture industries in South America. Some of our top clients include Molymet, Klabin, and Compañia Minera del Pacífico.
- Conceive, plan, and prioritize data projects.
- Align data projects with organizational goals.

Center for Applied Intelligent Systems Research — September 2020 – January 2021
Halmstad University, Halmstad, Sweden — (Visiting researcher)

- Created saliency maps for interpreting and explaining the decision process of deep neural networks to classify several types of dementia.

Center for Applied Intelligent Systems Research — September 2019 – November 2019
Halmstad University, Halmstad, Sweden — (Visiting researcher)

- Developed a deep learning model to distinguish between the metabolic process of Dementia with Lewy Bodies (DLB) and Cognitively Normal (CN) subjects of the largest database for this disease in Europe collected by the European DLB consortium.

Darmiyan — June 2019 – July 2019
San Francisco, California — (Data analyst)

- Developed deep convolutional neural networks for classifying neuroimaging data from Mild Cognitive Impaired (MCI) subjects who will develop into Alzheimer's disease or remain as MCI after a 5-year window.

PUBLICATIONS

- "Model compression using knowledge distillation with integrated gradients". *Neural Computing and Applications.* May 2026
- "Unsupervised skin feature tracking using deep neural networks". *Accepted on EURASIP Journal on Image and Video Processing.* July 2025
- "ChatGPT struggle to recognize reproducible science". *Knowledge and Information Systems.* April 2025. **doi:10.1007/s10115-025-02428-z**
- "Skin feature point tracking using deep feature encodings". *International Journal of Machine Learning and Cybernetics*, October 2024. **doi:10.1007/s13042-024-02405-y**
- "Age prediction using resting-state functional MRI". *Neuroinformatics*, February 2024. **doi:10.1007/s12021-024-09653-x**
- "A 3D deep learning model to predict the diagnosis of dementia with Lewy bodies, Alzheimer's disease, and mild cognitive impairment using brain 18F-FDG PET".*European Journal of Nuclear Medicine and Molecular Imaging*, Jan 2022. **doi:10.1007/s00259-021-05483-0**

- "Adopting Transfer Learning for Neuroimaging: a Comparative Analysis with a Custom 3D Convolution Neural Network Model". *BMC Medical Informatics & Decision Making*, November 2022. **doi:10.1186/s12911-022-02054-7**

CONFERENCES AND WORKSHOPS

- MadeAI2025: The 2nd International Conference of Modelling, Data Analytics and AI in Engineering with a paper titled "Knowledge Distillation: Enhancing Neural Network Compression with Integrated Gradients".
- International Union of Biochemistry and Molecular Biology Focused Meeting on Neurodegenerative Disease 2021 with poster "Meta-analysis of performance of existing machine learning models for detection of Alzheimer's disease".
- 4th Online Computer Vision and Artificial Intelligence Workshop, September 2021 with presentation "Facial feature point detection using deep feature encodings". The video is the highest viewed talk of the workshop with roughly 1200 views at the end of the workshop.

VOLUNTEERING

Secretaría Nacional de Ciencia, Tecnología e Innovación November 2022 - May 2024
Panama City, Panama (Reviewer of technical proposals)

- Evaluated technical proposals for the Public Calls for Innovative Entrepreneurship from 2022 to 2024 based on their technical and commercial feasibility.

SKILLS

- Deep learning: Artificial neural networks, Convolutional neural networks, Transformer networks, Time series analysis, Object classification, Object detection, Multivariate regression, Transfer learning, Autoencoders, Data visualization, Model explainability, Statistics
- Software: Matlab, C/C++, CUDA, Python, Keras, Tensorflow, MLflow, Latex, Scikit-learn, Pandas, Git
- Strategic planning, Project management, Cross-functional collaboration

HONORS AND CERTIFICATIONS

- Taiwan Employment Gold Card from 2024 to 2027.
- Taiwan International Cooperation and Development Fund (ICDF) scholarship from 2012 to 2016, which is awarded to outstanding students from countries with diplomatic relationships with Taiwan.
- Distinguished International Student Scholarship at National Cheng Kung University from 2017 to 2021.
- Hononary member of the Phi Tau Phi scholastic honor society since 2024. Awarded by National Cheng Kung University to the top 10% of doctoral graduands for excellent academic performance and moral conduct.

LANGUAGES

Spanish (Native), English (TOEFL: 110/120), Chinese (TOCFL: A2), French (DELF: A2)

REFERENCE LINKS

Nordlinglab profile, LinkedIn, Google Scholar